

Detecting Early Atrophy and Barcoding in OCT

An Interpretable, Reproducible Image-Analysis Pipeline

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The Question We Are Trying to Answer

An OCT B-scan is a cross-sectional image of the retina. We want to locate and measure two patterns beneath the retina:

Early atrophy (EA)

A broader region with increased transmission of light into deeper tissue.

Barcoding

Repeated vertical bright–dark changes that form a stripe-like pattern.

The final output is intentionally simple:

each horizontal column $\longrightarrow$ {**normal**, **EA**, **barcoding**}.

We then report the **number of detected intervals** and their **widths in pixels**. This is exploratory research software, not a clinically validated diagnostic system.

What Was Built

The project was reorganized into five independent stages:

Loading → Preprocessing → Detection → Evaluation → Visualization

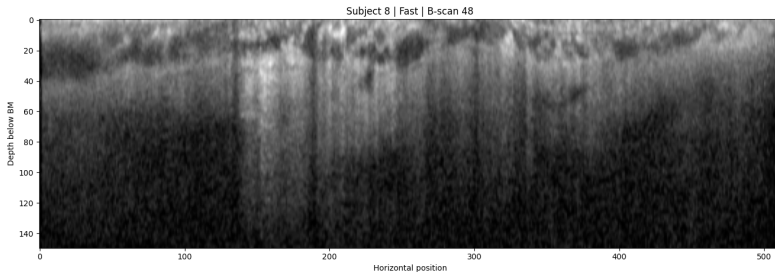
- **Loading:** accepts native HEYEX E2E volumes or PNG + JSON pairs.
- **Preprocessing:** aligns, crops, normalizes, and denoises scans.
- **Detection:** independently identifies EA and barcoding, then rejects likely anatomy and vessel shadows.
- **Evaluation:** compares automatic results with manual labels.
- **Visualization:** overlays detections and saves numerical JSON.

Each stage has a terminal script with an editable configuration. A separate validation script runs the complete workflow on all ten selected scans.

Why Preprocessing Is Necessary

Raw OCT scans vary in retinal curvature, brightness, depth, and speckle noise. The detector should respond to disease patterns rather than these acquisition differences.

- ① Use the supplied Bruch's membrane (BM) boundary as a landmark.
- ② Shift each image column so BM becomes horizontal.
- ③ Keep the same 150-pixel region beneath BM in every scan.
- ④ Put brightness on a comparable scale using z-score normalization.
- ⑤ Apply mild Gaussian denoising while preserving narrow vertical edges.



Selected Preprocessing Configuration

Setting	Selected value	Plain-language purpose
Reference layer	BM	Align every scan to the same anatomical boundary.
Reference row	Automatic median	Avoid imposing one scanner-specific row.
Depth below BM	150 pixels	Analyze a consistent deep-tissue region.
Include boundary	Yes	Retain the BM row at the top of the crop.
Require full depth	No	Pad a short crop instead of failing.
Normalization	Z-score	Express brightness relative to that scan's mean and spread.
Gaussian smoothing	(1.0, 0.5)	Smooth more through depth than horizontally, preserving stripes.
Fill values	0	Use the normalized center value for shifted or padded pixels.

These settings produced a consistent detector input of approximately 150×512 pixels.

What Normalization and Denoising Mean

Z-score normalization

For every pixel:

$$Z = \frac{\text{brightness} - \text{scan mean}}{\text{scan standard deviation}}.$$

A value near zero is typical for that scan; a positive value is brighter than typical. This reduces scan-to-scan brightness differences.

Mild Gaussian denoising

Nearby pixels are gently averaged. The selected strength is greater vertically (1.0) than horizontally (0.5), reducing granular OCT noise without smearing the narrow horizontal boundaries of vertical patterns.

The purpose is not to make the image look prettier. It is to make the same biological pattern produce more comparable measurements across patients.

How the Detector Reads a Scan

The detector summarizes every horizontal position using four main signals:

- **Typical column brightness:** is the deep tissue broadly bright?
- **Upper brightness:** how bright are the brightest relevant pixels?
- **Depth continuity:** does the pattern persist downward?
- **Vertical organization:** does the texture form organized columns?

Each signal is compared with the rest of the same scan using a robust median and interquartile range. Extreme values are clipped to $[-5, 5]$ so a single artifact cannot dominate the decision.

Eight calibrated inputs are used: the four signals above and four pairwise interactions. The interactions ask questions such as, “Is this region both bright *and* vertically continuous?”

Selected Feature-Extraction Settings

Measurement	Value	Plain-language effect
Verticality smoothing	1.0	Reduce speckle before measuring direction.
Verticality threshold	0.60	Require clear vertical organization.
Gradient quantile	0.80	Keep the strongest 20% of local edges.
Minimum component size	0	Do not remove structures by pixel area here.
Column brightness quantile	0.90	Summarize the brighter end of each column.
Minimum valid pixels	5	Avoid estimates based on too few pixels.
Column-signal smoothing	2.0	Suppress very short horizontal fluctuations.
Median / upper IQR multipliers	1.0 / 0.5	Set scan-relative brightness requirements.
Continuity window / depth lag	15 / 4	Test whether structure persists nearby and downward.
Continuity quantile	0.60	Retain above-typical continuity.
Vertical-fraction quantile	0.70	Retain the most vertically organized 30%.
Edge margin	10 columns	Ignore unreliable scan edges.

These values create the measurements supplied to the separate calibrated EA, barcoding, and structural-rejection models.

Separate EA and Barcoding Decisions

EA and barcoding are not treated as stages of the same rule. Two independently calibrated models assign a probability to every column:

Barcoding model

Selected when probability ≥ 0.55 .
Requires at least 8 consecutive columns and may bridge gaps up to 4 columns.

EA model

Selected when probability ≥ 0.50 .
Requires at least 12 consecutive columns and may bridge gaps up to 5 columns.

If both models select one column, the detector chooses the class whose probability is stronger relative to its own threshold. The first and last 10 columns are excluded to reduce edge artifacts.

V3: Requiring Spatial Context

Earlier versions often selected isolated anatomy that looked disease-like in one column. V3 requires neighboring evidence.

Context setting	Barcoding	EA	Meaning
Probability smoothing	2.0	2.5	Look at a small horizontal neighborhood.
Weight on original score	0.65	0.60	Preserve local evidence while adding context.
Support radius	5	7	Check 11 or 15 nearby columns.
Minimum nearby support	55%	60%	Most neighbors must show near-threshold evidence.
Required feature votes	2 of 3	3 of 3	EA uses stricter brightness, continuity, and verticality agreement.
Interval mean probability	0.52	0.48	Reject an entire weak interval.
Interval peak probability	0.68	0.62	Every retained interval needs a convincing core.

The less strict barcoding vote was retained because requiring all three features caused a large loss of true barcoding regions.

Rejecting Vessels and Structural Shadows

The manual annotations also identify **uncertain** and **vessel/structural** areas. These are excluded from performance scoring; the detector is not expected to label them as EA or barcoding.

- A calibrated structural-rejection model uses 11 intensity and texture measurements.
- A structural probability of at least 0.90 activates the veto.
- Structural runs must last 3 columns; a 1-column gap may be joined.
- The veto expands 2 columns on both sides to cover the shadow boundary.
- A second dark-shadow rule requires upper brightness ≤ -1.0 and continuity ≥ 0.25 ; it expands 3 columns on either side.

Disease gap-filling occurs *before* this veto, so rejected structural columns cannot accidentally be filled back into a disease interval.

How the Detector Was Tested

Ten B-scans were manually annotated, with a different selected scan for each patient:

- five fast-progressing subjects and five slow-progressing subjects;
- manual labels for normal, EA, barcoding, uncertain, and structural regions;
- automatic processing begins from the original E2E volume;
- uncertain and vessel/structural columns are ignored during scoring;
- EA and barcoding are evaluated separately, one column at a time.

This is a useful development set, but it is small and was also used to select parameters. The reported results measure current fit to these ten scans; they do not yet demonstrate performance on new patients.

What the Performance Metrics Mean

Precision

Of everything the detector marked, how much was correct?

High precision means fewer false alarms.

Recall

Of all manually marked disease, how much did the detector find?

High recall means fewer missed regions.

Dice score

One summary balancing overlap, false alarms, and misses.

0 = no overlap; 1 = perfect overlap.

Example: precision of 0.58 means about 58 of every 100 marked columns were correct. Recall of 0.52 means about 52 of every 100 manually marked columns were recovered.

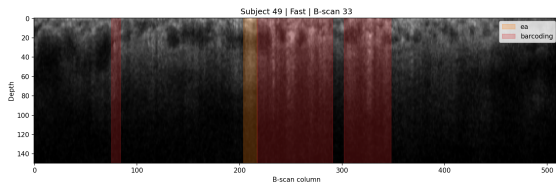
Best Current Aggregate Results

Class	Version	Precision	Recall	Dice
Barcoding	V2	0.526	0.506	0.516
Barcoding	V3 contextual	0.584	0.520	0.550
EA	V2	0.369	0.739	0.492
EA	V3 contextual	0.475	0.699	0.566

- Barcoding false-positive columns decreased from 371 to 302.
- EA false-positive columns decreased from 222 to 136.
- V3 trades a small amount of EA recall for substantially fewer false alarms and better overall overlap.

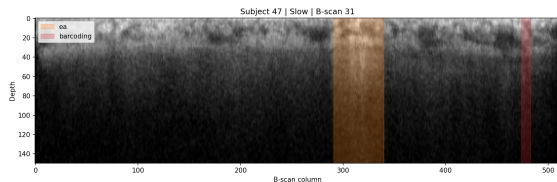
Interpretation: V3 is the best configuration tested so far, but an aggregate Dice near 0.55 is still development-stage performance.

Examples of Strong Current Results



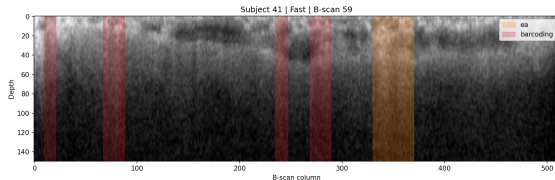
Barcoding: precision 0.930, recall 0.779, Dice 0.848

Orange denotes EA and red denotes barcoding. These examples show that the pipeline can closely localize both phenotypes on some scans.

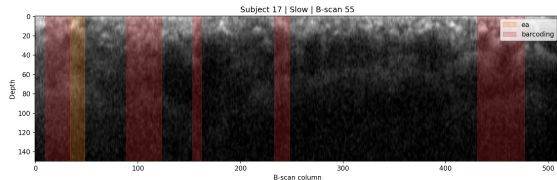


EA: precision 1.000, recall 0.907, Dice 0.951

Where the Detector Still Fails



Fast 41: unrelated structures survive as high-confidence detections.



Slow 17: a disease-negative scan still produces several intervals.

Spatial cleanup has improved precision, but normal anatomy can still resemble disease. The next major improvement should be an explicit **normal rejection model** using lesion-level depth profiles and independent validation patients.

Reproducible Workflow

After editing the configuration dictionaries in the corresponding scripts:

- ① Load a selected scan: `python scripts/load_data.py`
- ② Preprocess it: `python scripts/preprocess_data.py`
- ③ Run detection: `python scripts/run_detector.py`
- ④ Extract interval counts and widths: `python scripts/extract_numerical_results.py`
- ⑤ Create an overlay: `python scripts/visualize_detector.py`

For the ten annotated scans, one command runs the complete E2E workflow:

```
python scripts/validation_test.py
```

Outputs use matching names such as `fast_08_bscan_048_automatic.json` and `fast_08_bscan_048_automatic.png`.

- A complete, configuration-driven pipeline now loads, preprocesses, detects, measures, evaluates, and visualizes OCT findings.
- Consistent BM flattening, a 150-pixel crop, z-score normalization, and mild (1.0, 0.5) Gaussian denoising are the selected preprocessing settings.
- EA and barcoding use separate calibrated decisions, spatial context, whole-interval confidence, and structural-shadow rejection.
- Contextual V3 is the best tested detector: Dice 0.550 for barcoding and 0.566 for EA across the ten annotated scans.
- Performance varies substantially by patient. More annotations, a dedicated normal model, and testing on untouched patients are needed before clinical interpretation.

Questions?